

Students perform a synthesis of a carboxylic acid derivative.

They mix a 10.00 mL sample of hexanoic acid, 2.33 mL of ethanol, and 0.64 mL of concentrated sulfuric acid in a flask with a water condenser fitted to its neck. The mixture is boiled for 1 hr and then poured into water when the reaction is complete. The aqueous and organic layers are separated, and the organic layer is washed once with NaHCO_3 and twice with water. Solid Na_2SO_4 is added to the organic layer to remove any water present. The organic layer and Na_2SO_4 are separated by filtration. The filtrate is purified by fractional distillation to yield the carboxylic acid derivative, a colorless liquid.

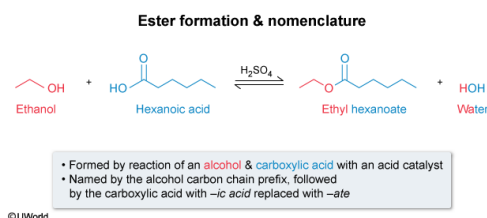
What is the product formed by the reaction of hexanoic acid and ethanol described in the passage?

- ☐ A. Acetic hexanoic anhydride [8%]
☐ B. Hexanoic ethanoic anhydride [11%]
☐ C. Hexyl ethanoate [17%]
✔ ☒ D. Ethyl hexanoate [62%]

Correct

61% answered correctly

Explanation:



Esters are **carboxylic acid derivatives** formed by **Fischer esterification**, an acid-catalyzed reaction of an **alcohol** with a **carboxylic acid**. In the first step of the reaction, the acid catalyst protonates the carbonyl oxygen of the carboxylic acid, making the carbonyl a good electrophile. Next, the alcohol acts as the nucleophile and attacks the carbonyl carbon. The carboxylic acid $-\text{OH}$ group then gets protonated and leaves as water, yielding an ester composed of the carbon chains contributed by the alcohol and the acid.

Esters are named by first stating the prefix that corresponds to the alcohol alkyl chain followed by the name of the carboxylic acid. The carboxylic acid **suffix** **-ic acid** is replaced by **-ate**. The passage states that ethanol and hexanoic acid underwent an acid-catalyzed reaction to form a carboxylic acid derivative. Therefore, the product formed was the ester **ethyl hexanoate**.

(Choices A and B) Anhydrides are formed by the condensation of two carboxylic acids rather than a carboxylic acid and an alcohol. Hexanoic acid would have to **react** with ethanoic acid (commonly known as acetic acid) to form ethanoic hexanoic anhydride (also known as acetic hexanoic anhydride).

(Choice C) Hexyl ethanoate is formed from ethanoic (acetic) acid and hexanol rather than hexanoic acid and ethanol.

Educational objective:

Esters are carboxylic acid derivatives formed by Fisher esterification, an acid-catalyzed reaction of a carboxylic acid and an alcohol. Esters are named by stating the alcohol chain prefix followed by the name of the carboxylic acid, with the suffix **-ic acid** replaced by **-ate**.

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Subject:
Organic Chemistry

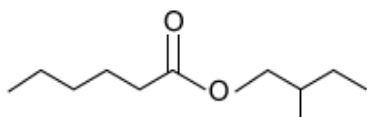
QID:402535
System:
Functional Groups and Their Reactions

Students perform a synthesis of a carboxylic acid derivative.

They mix a 10.00 mL sample of hexanoic acid, 2.33 mL of ethanol, and 0.64 mL of concentrated sulfuric acid in a flask with a water condenser fitted to its neck. The mixture is boiled for 1 hr and then poured into water when the reaction is complete. The aqueous and organic layers are separated, and the organic layer is washed once with NaHCO_3 and twice with water. Solid Na_2SO_4 is added to the organic layer to remove any water present. The organic layer and Na_2SO_4 are separated by filtration. The filtrate is purified by fractional distillation to yield the carboxylic acid derivative, a colorless liquid.

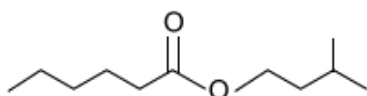
If the carboxylic acid derivative that formed in the passage is reacted with an excess of 3-methylbutan-1-ol and an acid catalyst, which of the following compounds will be the product?

☐ A.



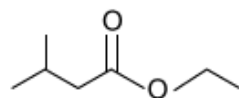
[15%]

☒ B.



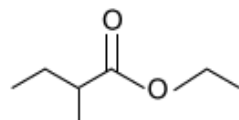
[74%]

☐ C.



[5%]

☐ D.

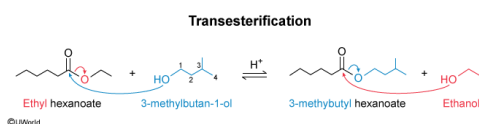


[3%]

Correct

73% answered correctly

Explanation:



Transesterification is a reaction between an ester and an alcohol in which the ester -OR group is replaced with the alcohol -OR group. Transesterifications can be either **acid-catalyzed** or **base-catalyzed**. The alcohol acts as a nucleophile and attacks the electrophilic carbonyl carbon of the ester, adding a new -OR group to the ester. The original -OR group leaves and is then protonated. Altogether, this process yields a new ester and a new alcohol.

Alcohols are named based on the longest continuous carbon chain containing the hydroxyl group (-OH). The **prefix** corresponds to the number of carbons in the alkyl chain, and the **suffix** -ol corresponds to the hydroxyl functional group.

The carboxylic acid derivative formed from the reaction described in the passage is the ester ethyl hexanoate, which contains a two-carbon chain derived from ethanol and a six-carbon chain derived from hexanoic acid. The alcohol 3-methylbutan-1-ol contains a four-carbon chain (butyl group) with a hydroxyl group (alcohol) on C-1 and a methyl group (-CH_3) on C-3. When ethyl hexanoate reacts with

3-methylbutan-1-ol, the ethyl group on the ester is replaced with the alcohol alkyl chain (3-methylbutyl), forming a new ester named 3-methylbutyl hexanoate, as shown in Choice B.

(Choice A) The methyl substituent on the alkyl group is at C-2 and would form if from a reaction with 2-methylbutan-1-ol rather than 3-methylbutan-1-ol. This placement of the methyl group could be obtained if the carbon chain were numbered starting at the incorrect end (ie, giving –OH group the highest number instead of the lowest).

(Choices C and D) These structures would result if the alkyl group adjacent to the *carbonyl* were replaced rather than the alkyl group adjacent to the ester oxygen.

Educational objective:

Transesterification is a reaction between an ester and an alcohol where the ester –OR group is replaced with the alcohol –OR group, forming a new ester and a new alcohol. Alcohols are named based on the longest carbon chain containing the hydroxyl group. The prefix corresponds to the number of carbons in the alkyl chain, and the suffix –*ol* corresponds to the hydroxyl functional group.

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Organic Chemistry

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System:
Functional Groups and Their Reactions

Students perform a synthesis of a carboxylic acid derivative.

They mix a 10.00 mL sample of hexanoic acid, 2.33 mL of ethanol, and 0.64 mL of concentrated sulfuric acid in a flask with a water condenser fitted to its neck. The mixture is boiled for 1 hr and then poured into water when the reaction is complete. The aqueous and organic layers are separated, and the organic layer is washed once with NaHCO_3 and twice with water. Solid Na_2SO_4 is added to the organic layer to remove any water present. The organic layer and Na_2SO_4 are separated by filtration. The filtrate is purified by fractional distillation to yield the carboxylic acid derivative, a colorless liquid.

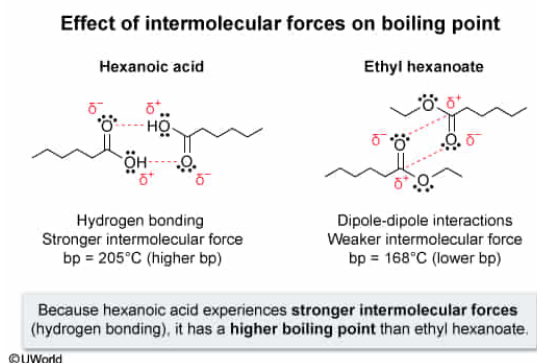
A majority of excess hexanoic acid is removed from the product in the extraction described in the passage; however, a small amount will remain, making the product impure. Which statement accurately describes the subsequent distillation of the product?

- ☐ A. Hexanoic acid will distill first because it has a lower boiling point than the product. [30%]
- ☐ B. Decreasing the pressure of the distillation will improve the purification of the product. [6%]
- ✓ ☒ C. The product will distill first because hexanoic acid has a higher boiling point. [61%]
- ☐ D. Increasing the length of the fractionating column will not give separation of the product and hexanoic acid. [2%]

Correct

61% answered correctly

Explanation:



Fractional distillation is a purification technique that separates molecules based on their **boiling point**, which is dependent on the intermolecular forces holding the molecules together. **Stronger intermolecular forces** require more energy to break, causing molecules with these forces to have a **higher boiling point** than molecules of similar molecular weight with weaker intermolecular forces.

Hydrogen bonds are strong intermolecular forces that consist of a **donor and acceptor**.

Donors are hydrogen atoms bonded to an electronegative atom from the second row of the

periodic table; acceptors are electronegative atoms with a lone pair of electrons (eg, oxygen, nitrogen, fluorine).

Hexanoic acid can **hydrogen bond** with **itself** (ie, to other hexanoic acid molecules) because it has a hydrogen bond donor (acidic hydrogen) and two acceptors (oxygen atoms). However, the reaction product **ethyl hexanoate** (an ester) is **only** a hydrogen bond acceptor, and therefore **cannot hydrogen bond with itself**. Esters experience dipole-dipole interactions, which are **weaker** than hydrogen bonds. Therefore, ethyl hexanoate has a **lower boiling point** and will **distill before hexanoic acid**.

(Choice A) Hexanoic acid can hydrogen bond with itself whereas ethyl hexanoate only experiences dipole-dipole interactions and cannot hydrogen bond with itself. Because hexanoic acid experiences stronger intermolecular forces than ethyl hexanoate, it will have a *higher* boiling point and distill *after* ethyl hexanoate.

(Choice B) Decreasing the pressure by distillation under vacuum would decrease the boiling point of *both* compounds and would not appreciably affect the difference between the boiling points.

(Choice D) Increasing the length of the column in a fractional distillation will *improve* the separation of the compounds.

Educational objective:

Boiling points are dependent on a molecule's intermolecular forces, and strong intermolecular forces lead to higher boiling points. Carboxylic acids contain a hydrogen bond donor (hydroxyl hydrogen) and hydrogen bond acceptors (carbonyl oxygen, hydroxyl oxygen), and therefore can hydrogen bond with themselves. Because hydrogen bonds are strong intermolecular forces, carboxylic acids have higher boiling points than molecules of a similar molecular weight that cannot hydrogen bond to themselves and have weaker intermolecular forces.

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Subject:
Organic Chemistry

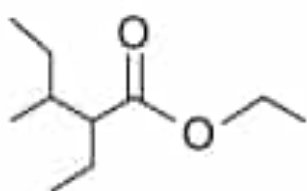
QID:402537

System:
Functional Groups and Their Reactions

Students perform a synthesis of a carboxylic acid derivative.

They mix a 10.00 mL sample of hexanoic acid, 2.33 mL of ethanol, and 0.64 mL of concentrated sulfuric acid in a flask with a water condenser fitted to its neck. The mixture is boiled for 1 hr and then poured into water when the reaction is complete. The aqueous and organic layers are separated, and the organic layer is washed once with NaHCO_3 and twice with water. Solid Na_2SO_4 is added to the organic layer to remove any water present. The organic layer and Na_2SO_4 are separated by filtration. The filtrate is purified by fractional distillation to yield the carboxylic acid derivative, a colorless liquid.

Which carboxylic acid would be needed to form Compound **A** (shown below) based on the reaction described in the passage?



Compound **A**

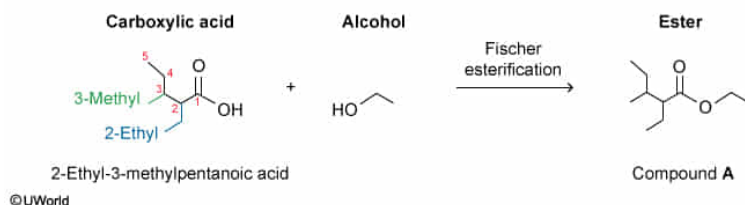
- ☐ A. 2-sec-butylbutanoic acid [2%]
- ✓ ☒ B. 2-ethyl-3-methylpentanoic acid [65%]
- ☐ C. 2,3-diethylbutanoic acid [22%]
- ☐ D. 4-ethyl-3-methylpentanoic acid [9%]

Correct

65% answered correctly

Explanation:

Fischer esterification & carboxylic acid nomenclature



Esters are formed by an acid-catalyzed reaction known as a **Fischer esterification**, in which a carboxylic acid and an alcohol combine. The alkyl group bonded to the carbonyl carbon comes from the carboxylic acid, and the alkyl chain bonded to the ester oxygen atom comes from the alcohol.

Carboxylic acids are named based on the longest continuous carbon chain **containing the carboxylic acid** functional group (known as the parent chain). The **parent chain is numbered beginning** with the **carbonyl** carbon of the carboxyl group. **Substituents** on the parent chain are named beginning with the parent chain **carbon number** to which it is bonded, followed by the **prefix** denoting the number of carbons in the substituent. The substituents are listed in alphabetical order. The parent chain is named last, and for carboxylic acids, the suffix **–e** of the parent alkane is replaced with **–oic acid**.

The parent chain of the carboxylic acid used to form Compound **A** has five carbons (prefix *pent–*). There are two substituents, an ethyl group on C-2 and a methyl group on C-3. Therefore, the carboxylic acid used is 2-ethyl-3-methylpentanoic acid.

(Choice A) The name 2-sec-butylbutanoic acid incorrectly identifies the parent chain as a four-carbon chain with a **sec-butyl** substituent on C-2.

(Choice C) The name 2,3-diethylbutanoic acid incorrectly identifies the parent chain as a four-carbon chain with two ethyl substituents, one on C-2 and one on C-3.

(Choice D) The name 4-ethyl-3-methylpentanoic acid correctly identifies the five-carbon parent chain. However, the parent chain is numbered incorrectly, giving the carbonyl carbon the highest number instead of the lowest, which affects the numbering of the substituents as well.

Educational objective:

Carboxylic acids are named by identifying the parent chain (ie, the longest continuous carbon chain containing the carboxyl group) and numbering the carbons beginning with the carbonyl carbon. The substituents are named beginning with the parent chain carbon number to which it is bonded, followed by the substituent prefix. The parent chain is named last, replacing the –e from the parent alkane with –*oic acid*.

Time spent:0 sec

Subject:

Organic Chemistry

QID:402538

System:

Functional Groups and Their Reactions

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Which of the following reactions will produce the same organic product whether performed on hexanoic acid or on the carboxylic acid derivative described in the passage?

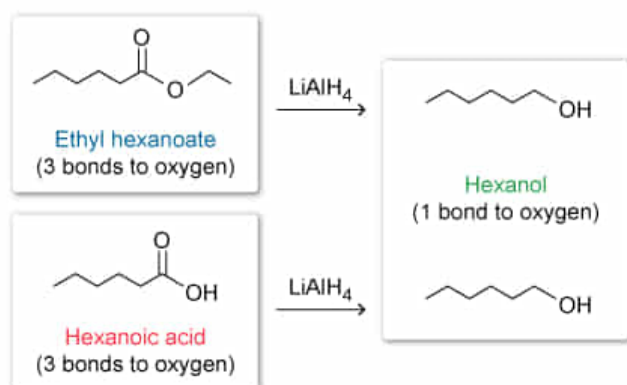
- ☐ A. Base-catalyzed aldol condensation [21%]
- ☐ B. Decarboxylation [15%]
- ☐ C. Oxidation with NaBH_4 [10%]
- ☒ D. Reduction with LiAlH_4 [51%]

Correct

52% answered correctly

Explanation:

Reduction with LiAlH_4



LiAlH_4 reduces carboxylic acids and esters to primary alcohols.

Reducing agents decrease an atom's **oxidation state**. The **reduction** of a carbon atom **decreases** the number of **bonds** it has to **electronegative atoms** (including oxygen, nitrogen, and halogens) and **increases** the number of **bonds** to less electronegative atoms (typically **hydrogen**). **Hydride**

reagents, such as LiAlH_4 and NaBH_4 , are common reducing agents and consist of a metal surrounded by one or more hydride ions. The hydride ion acts as a nucleophile and attacks an electrophilic atom. LiAlH_4 is a strong reducing agent and will **reduce carbonyl** compounds to **alcohols**. Aldehydes, esters, and carboxylic acids will be reduced to primary alcohols, and ketones will be reduced to secondary alcohols.

The reaction mixture contains either the ester ethyl hexanoate (ie, the derivative described in the passage) or hexanoic acid. If either of these two compounds are **reacted with LiAlH_4** , they would both be **reduced** to the six-carbon primary alcohol hexanol. Ethyl hexanoate will also produce ethanol as a side product when reduced with LiAlH_4 .

(Choice A) A **base-catalyzed aldol condensation** does require carbonyl compounds, but the required compounds are *ketones* and *aldehydes* rather than carboxylic acids and esters.

(Choice B) Decarboxylation occurs with **β -dicarbonyl** compounds. Neither ethyl hexanoate nor hexanoic acid contains a β -carbonyl, so they will not undergo decarboxylation.

(Choice C) NaBH_4 is a *reducing* agent rather than an **oxidizing agent**. Furthermore, NaBH_4 is a mild reducing agent and cannot reduce carboxylic acids or esters; therefore, hexanoic acid and ethyl hexanoate would not react with NaBH_4 .

Educational objective:

Reducing agents decrease an atom's oxidation state and reduce the number of bonds carbon has to electronegative atoms while increasing the number of bonds to less electronegative atoms (commonly hydrogen). Hydride reagents, such as LiAlH_4 and NaBH_4 , contain a metal atom surrounded by hydride ions. LiAlH_4 is a strong reducing agent and will reduce carbonyl

compounds, including esters and carboxylic acids, to alcohols.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402539

System:
Functional Groups and Their Reactions